

$$F.XS = \langle \max_i : 0 \leq i < \#XS : \text{Sum}(XS \uparrow i) + \text{Prod}(XS \downarrow i) \rangle \quad (H)$$

$$(CR) \quad XS = (XD XS)$$

$$\langle \max_i : 0 \leq i < \#(XD XS) : \text{Sum}((XD XS) \uparrow i) + \text{Prod}((XD XS) \downarrow i) \rangle$$

$\equiv \{ \text{Definición length} \}$

$$\langle \max_i : 0 \leq i < 1 + \#XS : \text{Sum}((XD XS) \uparrow i) + \text{Prod}((XD XS) \downarrow i) \rangle$$

$\equiv \{ \text{Partición de rango} \}$

$$(\text{Sum}((XD XS) \uparrow 0) + \text{Prod}((XD XS) \downarrow 0)) \max \langle \max_i : 0 \leq i < \#XS : \text{Sum}((XD XS) \uparrow (i+1)) + \text{Prod}((XD XS) \downarrow (i+1)) \rangle$$

$\equiv \{ \text{Def take y drop} \}$

$$(\text{Sum}.C] + \text{Prod}.(XD XS)) \max \langle \max_i : 0 \leq i < \#XS : \text{Sum}.(XD(XS \uparrow i)) + \text{Prod}(XS \downarrow i) \rangle$$

$\equiv \{ \text{Def sum} \}$

$$(\text{Sum}.C] + \text{Prod}.(XD XS)) \max \langle \max_i : 0 \leq i < \#XS : \text{X} + \text{Sum}(XS \uparrow i) + \text{Prod}(XS \downarrow i) \rangle$$

generalizo

$$F.XS.n \equiv \langle \max_i : 0 \leq i < \#XS.n + \text{Sum}(XS \uparrow i) + \text{Prod}(XS \downarrow i) \rangle \quad (H)$$

$$(CB) \quad F.C].n \equiv \langle \max_i : 0 \leq i < \#C].n + \text{Sum}(C] \uparrow i) + \text{Prod}(C] \downarrow i) \rangle$$

$\equiv \{ \text{Def length, logica} \}$

$$\langle \max_i : \text{false} : n + \text{Sum}(C] \uparrow i) + \text{Prod}(C] \downarrow i) \rangle$$

$\equiv \{ \text{Rango vacío} \}$

$\boxed{1-\infty}$

$$(CR) \quad XS = (XD XS)$$

$$\langle \max_i : 0 \leq i < \#(XD XS).n + \text{Sum}(XD XS \uparrow i) + \text{Prod}(XD XS \downarrow i) \rangle$$

$\equiv \{ \text{Definición length} \}$

$$\langle \max_i : 0 \leq i < 1 + \#XS : n + \text{Sum}((XD XS) \uparrow i) + \text{Prod}((XD XS) \downarrow i) \rangle$$

$\equiv \{ \text{Partición de rango} \}$

$$n + \text{Sum}((XD XS) \uparrow 0) + \text{Prod}((XD XS) \downarrow 0)) \max \langle \max_i : 0 \leq i < \#XS : n + \text{Sum}((XD XS) \uparrow (i+1)) + \text{Prod}((XD XS) \downarrow (i+1)) \rangle$$

$\equiv \{ \text{Def take y drop} \}$

$$(n + \text{Sum}.C] + \text{Prod}.(XD XS)) \max \langle \max_i : 0 \leq i < \#XS : n + \text{Sum}.(XD(XS \uparrow i)) + \text{Prod}(XS \downarrow i) \rangle$$

$\equiv \{ \text{Def sum y Prod} \}$

$$(n + (x \cdot \text{Prod}.xs)) \max < \text{Max } i: 0 \leq i < \#xs: n + \text{sum}(xD(xs \uparrow i)) + \text{Prod}(xs \downarrow i) >$$

$\equiv \{ \text{Def sum} \}$

$$(n + (x \cdot \text{Prod}.xs)) \max < \text{Max } i: 0 \leq i < \#xs: (n+x) + \text{sum}(xs \uparrow i) + \text{Prod}(xs \downarrow i) >$$

$\equiv \{ \text{Por HI} \}$

$$n + (x \cdot \text{Prod}.xs) \max f.xs.(n+x)$$

$$gf: [Num] \rightarrow Num \rightarrow Num$$

$$gf.C].n \equiv \text{minBound}$$

$$gf.(xDxs).n \equiv n + (x \cdot \text{Prod}.xs) \max f.xs.(n+x)$$

$$\textcircled{2} \text{ sumInt}.xs \equiv < \text{Min}, as, bs, cs: xs = as ++ bs ++ cs : \text{sum}.bs > \textcircled{H}$$

$\textcircled{CR}$

$$< \text{Min}, as, bs, cs: (xDxs) = as ++ bs ++ cs : \text{sum}.bs >$$

$\equiv \{ \text{logical} \}$

$$< \text{Min}, as, bs, cs: (xDxs) = as ++ bs ++ cs \wedge (as = [] \vee as \neq []) : \text{sum}.bs >$$

$\equiv \{ \text{Distributividad, Partición de rango} \}$

$$\textcircled{1} < \text{Min}, as, bs, cs: (xDxs) = as ++ bs ++ cs \wedge as = [] : \text{sum}.bs > \text{min} \textcircled{H}$$

$$< \text{Min}, as, bs, cs: (xDxs) = as ++ bs ++ cs \wedge as \neq [] : \text{sum}.bs >$$

$$\textcircled{1} < \text{Min}, as, bs, cs: (xDxs) = as ++ bs ++ cs \wedge as = [] : \text{sum}.bs >$$

$\equiv \{ \text{Definición as, Def concatenar} \}$

$$< \text{Min}, bs, cs: (xDxs) = bs ++ cs : \text{sum}.bs >$$

$\textcircled{1} \text{ Modularización.}$

$$\text{mSumInt}.xs \equiv < \text{Min}, bs, cs: (xDxs) = bs ++ cs : \text{sum}.bs > \textcircled{H}$$

$$\textcircled{CB} \text{ mMinInt}.C] \equiv < \text{Min}, bs, cs: C] = bs ++ cs : \text{sum}.bs >$$

$\equiv \{ \text{Def concatenar} \}$

$$< \text{Min}, bs, cs: bs = [] \wedge cs = C] : \text{sum}.bs >$$

$\Rightarrow \text{eliminación de variable } cs, \text{ rango unitario}$

$$\text{sum}.C] >$$

$\equiv \{ \text{Def sum} \}$

$\textcircled{10}$

$$\textcircled{CR} msumIn.(x \text{D} x_s) \equiv \langle \text{Min}_{bs,cs} : (x \text{D} x_s) = bs ++ cs : \text{sum}.bs \rangle$$

$\equiv \{ \text{logica} \}$

$$\langle \text{Min}_{bs,cs} : (x \text{D} x_s) = bs ++ cs \wedge (bs = [] \vee bs \neq []) : \text{sum}.bs \rangle$$

$\equiv \{ \text{Distributividad del } \wedge \text{ con el } \vee, \text{ Partición de rango} \}$

$$\textcircled{1} \langle \text{Min}_{bs,cs} : (x \text{D} x_s) = bs ++ cs \wedge bs = [] : \text{sum}.bs \rangle \text{ min}$$

$$\textcircled{2} \langle \text{Min}_{bs,cs} : (x \text{D} x_s) = bs ++ cs \wedge bs \neq [] : \text{sum}.bs \rangle$$

$$\textcircled{1} \langle \text{Min}_{bs,cs} : (x \text{D} x_s) = bs ++ cs \wedge bs = [] : \text{sum}.bs \rangle$$

$\equiv \{ \text{Def } bs, cs, \text{ Def concatenar} \}$

$$\langle \text{Min}_{bs,cs} : (x \text{D} x_s) = cs : \text{sum}.[] \rangle$$

$\equiv \{ \text{Def sum, término constante} \}$

$$\textcircled{2} \text{ o min. } \langle \text{Min}_{bs,cs} : (x \text{D} x_s) = bs ++ cs \wedge bs \neq [] : \text{sum}.bs \rangle$$

$\equiv \{ \text{Cambio de Variable} \}$

$$\text{ o min. } \langle \text{Min}_{bs,cs} : (x \text{D} x_s) = (b \text{D} bs) ++ cs : \text{sum}.(b \text{D} bs) \rangle$$

$\equiv \{ \text{Def concatenar, Def sum} \}$

$$\text{ o min. } \langle \text{Min}_{bs,cs} : (x \text{D} x_s) = b \text{D} (bs ++ cs) : b + \text{sum}.bs \rangle$$

$\equiv \{ \text{Prop de listas} \}$

$$\text{ o min. } \langle \text{Min}_{bs,cs} : x = b \wedge x_s = bs ++ cs : b + \text{sum}.bs \rangle$$

$\equiv \{ \text{eliminación de variable} \}$

$$\text{ o min. } \langle \text{Min}_{bs,cs} : x_s = bs ++ cs : x + \text{sum}.bs \rangle$$

generalización $\downarrow$

$$gmsumIn.xs.n \equiv \langle \text{Min}_{bs,cs} : x_s = bs ++ cs : n + \text{sum}.bs \rangle \quad \textcircled{H}$$

$$\textcircled{CB} msumIn.[]n \equiv \langle \text{Min}_{bs,cs} : [] = bs ++ cs : n + \text{sum}.bs \rangle$$

$\equiv \{ \text{Def concatenar} \}$

$$\langle \text{Min}_{bs,cs} : bs = [] \wedge cs = [] : n + \text{sum}.bs \rangle$$

$\equiv \{ \text{eliminación var } cs \text{ y rango unitario} \}$

$$n + \text{sum}.[]$$

$\equiv \{ \text{Def sum} \}$ $\boxed{n}$

$$\textcircled{CR} gmsumIn.(x \text{D} x_s).n \equiv \langle \text{Min}_{bs,cs} : (x \text{D} x_s) = bs ++ cs : n + \text{sum}.bs \rangle$$

$\equiv \{ \text{logica} \}$

$$\langle \text{Min}_{bs,cs} : (X \text{ DXS}) = bs + tcs \wedge (bs = [] \vee bs \neq []) : n + \text{sum} - bs \rangle$$

$\equiv \{ \text{Distributividad, Partición de rango} \}$

$$\langle \text{Min}_{bs,cs} : (X \text{ DXS}) = bs + tcs \wedge bs = [] : n + \text{sum} - bs \rangle \text{ min} \quad \textcircled{1}$$

$$\langle \text{Min}_{bs,cs} : (X \text{ DXS}) = bs + tcs \wedge bs \neq [] : n + \text{sum} - bs \rangle \quad \textcircled{2}$$

$$\textcircled{2} \langle \text{Min}_{bs,cs} : (X \text{ DXS}) = bs + tcs \wedge bs \neq [] : n + \text{sum} - bs \rangle$$

$\equiv \{ \text{Def lista, Def Concat} \}$

$$\langle \text{Min}_{bs,cs} : (X \text{ DXS}) = cs : n + \text{sum} - [] \rangle$$

$\equiv \{ \text{Def sum, término constante} \}$

$$n. \text{ min} \textcircled{2} \langle \text{Min}_{bs,cs} : (X \text{ DXS}) = bs + tcs \wedge bs \neq [] : n + \text{sum} - bs \rangle$$

$\equiv \{ \text{Cambio de variable } bs \leftarrow (b \text{ bs}), \text{ Def Concat} \}$

$$n. \text{ min} \langle \text{Min}_{bs,cs} : (X \text{ DXS}) = b \text{ } (bs + tcs) : n + \text{sum}(b \text{ bs}) \rangle$$

$\equiv \{ \text{Def listas, Def sum} \}$

$$n. \text{ min} \langle \text{Min}_{bs,cs} : x = b \wedge xs = bs + tcs : n + b + \text{sum} - bs \rangle$$

$\equiv \{ \text{eliminación de variable, HI} \}$

$$\boxed{n. \text{ min } gmsumIn.xs.(n+x)} \leftarrow$$

$$(n \text{ min } gmsumIn.xs.(n+x)) \text{ min} \textcircled{2} \langle \text{Min}_{as,bs,cs} : (X \text{ DXS}) = as + tbs + tcs \wedge as \neq [] : \text{sum} - bs \rangle$$

$\equiv \{ \text{lógica, Def Concat, Def listas} \}$

$$(n \text{ min } gmsumIn.xs.(n+x)) \text{ min} \rightarrow \langle \text{Min}_{as,bs,cs} : x = a \wedge xs = as + tbs + tcs : \text{sum} - bs \rangle$$

$\equiv \{ \text{eliminación de variable, HI} \}$

$$\boxed{(n \text{ min } gmsumIn.xs.(n+x)) \text{ min } \text{sumIn}.xs}$$

$$\text{pf. } E.xs = \langle \text{Max } i : 0 \leq i < \#(ys) \wedge E + \text{sum}((ys) \upharpoonright i) = \text{sum}((ys) \upharpoonright xs) : i \rangle$$

$$\text{PF. } E.xs = \langle \text{Max } i : 0 \leq i < \#(\overline{ys}) \wedge E + \text{sum}((ys) \upharpoonright i) = \text{sum}((ys) \downarrow i) : i \rangle \quad \textcircled{H}$$

$\textcircled{CR}$

$$\text{PF. } E.(X \text{ DXS}) = \langle \text{Max } i : 0 \leq i < \#(\overline{ys(X \text{ DXS})}) \wedge (E + \text{sum}((ys(X \text{ DXS}) \upharpoonright i)) = \text{sum}((ys(X \text{ DXS}) \downarrow i)) : i \rangle$$

$\equiv \{ \text{Def length, lógica, Distributividad del } \wedge \text{ con } \vee \}$

$$\langle \text{Max } i : i = 0 \wedge E + \text{sum}((ys(X \text{ DXS}) \upharpoonright i)) = \text{sum}((ys(X \text{ DXS}) \downarrow i)) : i \rangle \text{ max} \quad \textcircled{1}$$

$$\langle \text{Max. } i: 2 \leq i < \#(xs) \wedge E + \text{Sum}((YD(xs)) \uparrow i) = \text{Sum}((YD(xs)) \downarrow i) : i \rangle^{(2)}$$

$\equiv \{ \text{Cambio de variable } i \leftarrow i+1; \text{aritmética} \}$

$$\langle \text{Max. } i: 0 \leq i < \#(xs) \wedge E + \text{Sum}((YD(xs)) \uparrow (i+1)) = \text{Sum}((YD(xs)) \downarrow (i+1)) : i+1 \rangle$$

$\equiv \{ \text{Def take, Drop} \}$

$$\langle \text{Max. } i: 0 \leq i < \#(xs) \wedge E + \text{Sum}(YD(\overset{xs}{(xs)} \uparrow i)) = \text{Sum}((xs) \downarrow i) : i+1 \rangle$$

$\equiv \{ \text{HI, Distributividad} \}$

$$\langle \text{Max. } i: i=0 \wedge E + \text{Sum}((YD(xs)) \uparrow i) = \text{Sum}((YD(xs)) \downarrow i) : i \rangle \max (PF.E.xs + 1)$$

$\equiv \{ \text{Aplicación rango al término} \}$

$$\langle \text{Max. } i: E + \text{Sum}((xs) \uparrow 0) = \text{Sum}((xs) \downarrow 0) : 0 \rangle$$

$\equiv \{ \text{Término constante} \}$

$$\boxed{0 \max (PF.E.xs + 1)}$$

$$(CB) PF.E.[] = \langle \text{Max } i: 0 \leq i < \#(YD[]) \wedge E + \text{Sum}(YD[] \uparrow i) = \text{Sum}(YD[] \downarrow i) : i \rangle^1$$

$\equiv \{ \text{Def length} \}$

$$\langle \text{Max } i: 0 \leq i < 1 \wedge E + \text{Sum}(YD[] \uparrow i) = \text{Sum}(YD[] \downarrow i) : i \rangle^1$$

$\equiv \{ \text{lógica, aplicación del rango al término} \}$

$$\langle \text{Max. } i: i=0 \wedge E + \text{Sum}(YD[] \uparrow 0) = \text{Sum}(YD[] \downarrow 0) : 0 \rangle^1$$

$\equiv \{ \text{Término constante} \}$

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$$\begin{aligned} PF : \text{Num} &\rightarrow [\text{Num}] \rightarrow \text{Num} \\ PF.E.[] &\doteq 0 \\ PF.E.(xs) &\doteq 0 \max (PF.E.xs + 1) \end{aligned}$$

1. a) Derivar una definición recursiva para la función especificada por:

$$f.xs = \langle \forall i: 0 \leq i < \#xs : (-1)^{xs!i} \geq 0 \rangle$$

a donde xs es una lista de números naturales (≥ 0).

b) Dar una lista xs de al menos 3 elementos que cumpla $f.xs = \text{True}$. Justificar.

c) Dar otra de al menos 3 elementos que cumpla $f.xs = \text{False}$. Justificar.

$$f.xs \doteq \langle \forall i: 0 \leq i < \#xs : (-1)^{xs!i} \geq 0 \rangle^{(H)}$$

a) (CP)

$$f(xs) \doteq \langle \forall i: 0 \leq i < \#(xs) : (-1)^{(xs)!i} \geq 0 \rangle$$

$\equiv \{ \text{lógica, Partición de rango} \}$

$$\langle \forall i: i=0 : (-1)^{(xs)!i} \geq 0 \rangle^{(2)} \wedge \langle \forall i: 1 \leq i < \#(xs) : (-1)^{(xs)!i} \geq 0 \rangle^{(2)}$$

$\equiv \{ \text{Cambio de variable } i \leftarrow i+1, \text{Def length, aritmética} \}$

$$\langle \forall i: 0 \leq i < \text{xs} : (-1)^{(\text{xs})!i} \geq 0 \rangle$$

$\equiv \{ \text{Def tirar} \}$

$$\langle \forall i: i=0 : (-1)^{(\text{xs})!i} \geq 0 \rangle \wedge \langle \forall i: 0 \leq i < \text{xs} : (-1)^{\text{xs}!i} \geq 0 \rangle$$

$\equiv \{ \text{HI} \}$

$$\langle \forall i: i=0 : (-1)^{(\text{xs})!i} \geq 0 \rangle \wedge \text{f.xs}$$

$\equiv \{ \text{Aplicación del rango al término} \}$

$$(-1)^{(\text{xs})!0} \geq 0 \wedge \text{f.xs}$$

$\equiv \{ \text{Def Prop} \}$

$$(-1)^n \geq 0 \wedge \text{f.xs}$$

$$\Rightarrow \boxed{\text{exp}(\text{xs}).n \cdot x \geq 0 \wedge \text{f.xs}}$$

• Modularizo

$$\text{exp} x.n \equiv x^n$$

$$\text{CB } \text{exp} x.n \equiv x^0 = \boxed{1}$$

$$\text{CR } \text{exp} x.(n+1) \equiv x^{n+1}$$

$\equiv \{ \text{Prop de Potencia} \}$

$$(x)^n \cdot x$$

$\equiv \{ \text{HI} \}$

$$\boxed{\text{exp} x.n \cdot x}$$

$$\text{CB } \text{f}(\text{xs}) \equiv \langle \forall i: 0 \leq i < \text{xs} : (-1)^{i!} \geq 0 \rangle$$

$\equiv \{ \text{rango vacío para el } \forall \}$

$$\boxed{\text{True}}$$

$$\text{f} : [\text{Num}] \rightarrow \text{Bool}$$

$$\text{f}(\text{xs}) \equiv \text{True}$$

$$\text{f}(\text{xs}) \equiv \text{exp} x.n \cdot x \wedge \text{f.xs}$$

b) Dar una lista xs de al menos 3 elementos que cumpla $\text{f.xs} = \text{True}$. Justificar.

c) Dar otra de al menos 3 elementos que cumpla $\text{f.xs} = \text{False}$. Justificar.

$$\text{Pares} = [0, 2, 4]$$

$$\text{b) } \text{f.xs} \equiv \langle \forall i: 0 \leq i < \text{xs} : (-1)^{\text{xs}!i} \geq 0 \rangle$$

$$\text{f}([0, 2, 4]) \equiv \langle \forall i: i \in \{0, 2, 4\} : (-1)^{[0, 2, 4]!i} \geq 0 \rangle$$

$\equiv \{ \text{Aplico rango al término} \}$

$$(-1)^0 \geq 0 \wedge (-1)^2 \geq 0 \wedge (-1)^4 \geq 0$$

$\equiv \{ \text{Por Def Potencia y lógica} \}$

$$\text{True} \wedge \text{True} \wedge \text{True}$$

$\equiv \{ \text{lógica} \}$

$$\boxed{\text{True}}$$

$$\text{c) } \text{xs} = [1, 2, 4]$$

$$\text{f.xs} \equiv \langle \forall i: 0 \leq i < \text{xs} : (-1)^{\text{xs}!i} \geq 0 \rangle$$

$$F.[1,2,1] = \langle \forall i: i \in \{0,1,2\} : (-1)^{[1,2,1]!i} \geq 0 \rangle$$

$\equiv \{ \text{Aplico rango al termino, indexar} \}$

$$(-1)^2 \geq 0 \wedge (-1)^2 \geq 0 \wedge (-1)^1 \geq 0$$

$\equiv \{ \text{Por Def Potencia y logica?} \}$

$$\text{False} \wedge \text{True} \wedge \text{True}$$

$\equiv \{ \text{logica} \}$

False.

2. a) Derivar una definición recursiva para la función especificada por:

$$\text{pin8}.xs = \langle \exists as, bs : xs = as ++ bs : \text{prod}.as = 8 \rangle$$

a donde *prod* es la función que calcula el producto de los elementos de una lista.

b) Dar una lista *xs* de al menos 3 elementos que cumpla $\text{pin8}.xs = \text{True}$. Justificar.

c) Dar otra de al menos 3 elementos que cumpla $\text{pin8}.xs = \text{False}$. Justificar.

$$\text{Pin8}.xs = \langle \exists as, bs : xs = as ++ bs : \text{Prod}.as = 8 \rangle \quad (H)$$

$$(R) \text{Pin8}.(\lambda xs) = \langle \exists as, bs : (\lambda xs) = as ++ bs : \text{Prod}.as = 8 \rangle$$

$\equiv \{ \text{logica} \}$

$$\langle \exists as, bs : (\lambda xs) = as ++ bs \wedge (as = [] \vee as \neq []) : \text{Prod}.as = 8 \rangle$$

$\equiv \{ \text{Distributividad y Partición de rango} \}$

$$\langle \exists as, bs : (\lambda xs) = as ++ bs \wedge as = [] : \text{Prod}.as = 8 \rangle \quad (1) \quad \vee \quad (2) \quad \langle \exists as, bs : (\lambda xs) = as ++ bs \wedge as \neq [] : \text{Prod}.as = 8 \rangle$$

$$(2) \langle \exists as, bs : (\lambda xs) = as ++ bs \wedge as \neq [] : \text{Prod}.as = 8 \rangle$$

$\equiv \{ \text{logica } as \neq [] \wedge as \in (as), \text{Def Concat} \}$

$$\langle \exists as, bs : (\lambda xs) = a :: (as ++ bs) : \text{Prod}.as = 8 \rangle$$

$\equiv \{ \text{Def listas, Def Product} \}$

$$\langle \exists as, bs : \underline{x} = a \wedge xs = as ++ bs : a \cdot \text{Prod}.as = 8 \rangle$$

$\equiv \{ \text{eliminación de variable} \}$

$$\langle \exists as, bs : xs = as ++ bs : \underline{x} \cdot \text{Prod}.as = 8 \rangle$$

$\equiv \{ \text{No llego a la HI, debo generalizar?} \}$

$$\text{gPin8}.xs.n = \langle \exists as, bs : xs = as ++ bs : n \cdot \text{Prod}.as = 8 \rangle \quad (H)$$

$$(R) \text{gPin8}.[] . n = \langle \exists as, bs : \underline{[]} = as ++ bs : n \cdot \text{Prod}.as = 8 \rangle$$

$\equiv \{ \text{Def concatenar} \}$

$$\langle \exists as, bs : as = [] \wedge bs = [] : n \cdot \text{Prod } as = 8 \rangle$$

$\equiv \{ \text{eliminación variable } bs, \text{ rango unitario} \}$

$$\langle \exists as : n \cdot \text{Prod } [] = 8 \rangle$$

$\equiv \{ \text{Def Prod, término constante} \}$

$$\boxed{n = 8}$$

$$\textcircled{C} \text{gPin8}.(xs) = \langle \exists as, bs : (xs) = as ++ bs : n \cdot \text{Prod } as = 8 \rangle$$

$\equiv \{ \text{lógica} \}$

$$\langle \exists as, bs : (xs) = as ++ bs \wedge (as = [] \vee as \neq []) : n \cdot \text{Prod } as = 8 \rangle$$

$\equiv \{ \text{Distributividad y Partición de rango} \}$

$$\langle \exists as, bs : (xs) = as ++ bs \wedge as = [] : n \cdot \text{Prod } as = 8 \rangle^{(1)} \vee \langle \exists as, bs : (xs) = as ++ bs \wedge as \neq [] : n \cdot \text{Prod } as = 8 \rangle^{(2)}$$

$$\textcircled{2} \langle \exists as, bs : (xs) = as ++ bs \wedge as \neq [] : n \cdot \text{Prod } as = 8 \rangle$$

$\equiv \{ \text{lógica } as \neq [], \text{Def concat} \}$

$$\langle \exists as, bs : (xs) = as ++ bs : n \cdot \text{Prod } (as ++ bs) = 8 \rangle$$

$\equiv \{ \text{Def listas, Def Product, Asociatividad} \}$

$$\langle \exists as, bs : x = a \wedge xs = as ++ bs : (n - 1) \cdot \text{Prod } as = 8 \rangle$$

$\equiv \{ \text{eliminación de variable} \}$

$$\langle \exists as, bs : xs = as ++ bs : (n - 1) \cdot \text{Prod } as = 8 \rangle$$

$\equiv \{ \text{HI} \}$

$$\langle \exists as, bs : (xs) = as ++ bs \wedge as = [] : n \cdot \text{Prod } as = 8 \rangle^{(1)} \vee \text{gPin8} \cdot xs \cdot (n - 1)$$

$\equiv \{ \text{Def de as, Def concatenar} \}$

$$\langle \exists bs : (xs) = bs : n \cdot \text{Prod } [] = 8 \rangle \vee \text{gPin8} \cdot xs \cdot (n - 1)$$

$\equiv \{ \text{Def Prod, término constante} \}$

$$\boxed{n = 8 \vee \text{gPin8} \cdot xs \cdot (n - 1)}$$

$$\bullet \text{gPin8} \cdot xs \cdot 1 = \text{Pin8} \cdot xs$$

$$\text{gPin8} : [\text{Num}] \rightarrow \text{Num} \rightarrow \text{Bool}$$

$$\text{gPin8} \cdot [] \cdot n \doteq (n = 8)$$

$$\text{gPin8} \cdot (xs) \cdot n \doteq (n = 8 \vee \text{gPin8} \cdot xs \cdot (n - 1))$$

b) Dar una lista xs de al menos 3 elementos que cumpla $\text{pin8} \cdot xs = \text{True}$. Justificar.

c) Dar otra de al menos 3 elementos que cumpla $\text{pin8} \cdot xs = \text{False}$. Justificar.

$$b) xs = [2, 2, 2]$$

$$\text{gPin8} \cdot xs \cdot n = \langle \exists as, bs : xs = as ++ bs : 2 \cdot \text{Prod } as = 8 \rangle$$

as	bs
[2, 2, 2]	[]
[2, 2]	[2]
[2]	[2, 2]

$$\text{giving } x.n \leq \exists a, b : \underline{[2, 2, 2] = a + b} : 1. \text{Prod. } a = 0 \quad [7] \mid [2, 2, 2]$$

$$\equiv \{ \text{Def } a \}$$

$$\langle \exists a, b : a \in \{[2, 2, 2], [2, 2], [2], []\} : 2 \text{Prod. } a = 0 \rangle$$

$$\equiv \{ \text{Aplicación rango al término} \}$$

$$\text{Prod}([2, 2, 2]) = 0 \vee \text{Prod}([2, 2]) = 0 \vee \text{Prod}([2]) = 0 \vee \text{Prod}([]) = 0$$

$$\equiv \{ \text{Def de producto, lógicos} \}$$

$$\text{True} \vee \text{False} \vee \text{False} \vee \text{False} \Rightarrow \boxed{\text{True}}$$

$$c) \quad xs = [1, 2, 2]$$

$$\text{giving } x.n = \langle \exists a, b : xs = a + b : 2. \text{Prod. } a = 0 \rangle$$

$$\text{giving } x.n \leq \exists a, b : \underline{[1, 2, 2] = a + b} : 1. \text{Prod. } a = 0$$

$$\equiv \{ \text{Def } a \}$$

$$\langle \exists a, b : a \in \{[1, 2, 2], [2, 2], [2], []\} : 2 \text{Prod. } a = 0 \rangle$$

$$\equiv \{ \text{Aplicación rango al término} \}$$

$$\text{Prod}([1, 2, 2]) = 0 \vee \text{Prod}([2, 2]) = 0 \vee \text{Prod}([2]) = 0 \vee \text{Prod}([]) = 0$$

$$\equiv \{ \text{Def de producto, lógicos} \}$$

$$\text{False} \vee \text{False} \vee \text{False} \vee \text{False} \Rightarrow \boxed{\text{False}}$$

3. Especificar funciones para resolver los siguientes problemas. También dar el tipo. No derivar.

✓ a) Calcular la cantidad de divisores que tiene un número $n > 0$ (sin contar el 1 ni n mismo).

b) Dadas dos listas xs e ys , calcular si xs tiene algún segmento cuya suma es mayor que la suma de ys . Ejemplo: con $xs = [-1, 8, -2, 4, -2]$, $ys = [3, 6]$ el resultado es afirmativo, por el segmento $[8, -2, 4]$ de xs .

$$a) \quad \langle \sum i : 2 \leq i \leq n : n \bmod i = 0 \rangle$$

$$b) \quad \langle \exists as, bs, cs, ds : xs = as ++ bs \wedge ys = cs ++ ds : (\text{sum } as > \text{sum } ys) \vee (\text{sum } bs > \text{sum } ys) \rangle$$

→ Dudo